JIAN WANG

Ph.D., conferred June 2026 | Software Engineering · LLMs · Trustworthy AI

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RESEARCH INTERESTS

Jian Wang's research centers on the trustworthy use of AI in software engineering, employing large language models (LLMs) and retrieval-augmented methods as the core engine to pursue three mutually reinforcing thrusts: AI-generated code detection, automated program repair, and execution-trace-grounded program understanding. His work has appeared at top-tier venues including ASE, ISSRE, EMNLP, ICML, NeurIPS, IJCAI, and TOSEM. The program aims to advance ***Trustworthy Code Intelligence*** for the *security*, *reliability*, *and sustainability* of modern software systems, with particular focus on safety-critical and continuously evolving ecosystems where AI-generated code introduces novel risks. Building on this foundation, his next-phase research will extend to a frontier challenge in AI for software engineering—***Long-Horizon Reasoning***: enabling AI to autonomously plan, iterate, and repair across multi-file, multi-step, dependency-heavy real-world maintenance scenarios. He aspires to build a research and teaching program at the intersection of **software engineering and trustworthy AI**, training the next generation of researchers who can both leverage and rigorously evaluate AI-assisted development.

EDUCATION

Ph.D. in Computer Science, Nanyang Technological University (NTU), Singapore
Aug. 2021 - June 2026

Advisor: *Prof. Yi Li* | Dissertation: *From Detection to Repair — A Framework for Reliable AI-Assisted Programming with LLMs*

B.Eng. in Software Engineering, Tianjin University, China *2007 – 2011*

PUBLICATIONS

Note: Self in bold; "*" = first author. Brackets at end = CCF tier / CORE ranking.

Publication summary: CCF-A **7 papers** (2 first-author: ASE'25, ASE'24); CCF-B **3 papers** (2 first-author: EMNLP-Findings'25, ISSRE'24); **4 first-author papers** spanning three mutually reinforcing thrusts: *AI-Code Detection → Post-Detection Program Repair → Execution Semantics for Hard-to-Repair Cases*.

Peer-Reviewed Conference Papers

[1] **Jian Wang***, Xiaofei Xie, Qiang Hu, Shangqing Liu, Jiongchi Yu, Jiaolong Kong, and Yi Li. "Defects4C: Benchmarking Large Language Model Repair Capability with C/C++ Bugs." *Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE)* , 2025. [CCF-A / CORE A*]

[2] **Jian Wang***, Xiaofei Xie, Qiang Hu, Shangqing Liu, and Yi Li. "Do Code Semantics Help? A Comprehensive Study on Execution Trace-Based Information for Code Large Language Models." *Findings of the Association for Computational Linguistics: EMNLP* , 2025. [CCF-B / CORE A*]

[3] **Jian Wang***, Shangqing Liu, Xiaofei Xie, and Yi Li. "An Empirical Study to Evaluate AIGC Detectors on Code Content." *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering (ASE)* , 2024, pp. 844–856. [CCF-A / CORE A*]

[4] **Jian Wang***, Shangqing Liu, Xiaofei Xie, Jing Kai Siow, Kui Liu, and Yi Li. "RATCHET: Retrieval Augmented Transformer for Program Repair." *Proceedings of the 35th IEEE International Symposium on Software Reliability Engineering (ISSRE)* , 2024, pp. 427–438. [CCF-B / CORE A]

[5] Shangqing Liu, Wei Ma**, Jian Wang**, Xiaofei Xie, Ruitao Feng, and Yang Liu. "Enhancing Code Vulnerability Detection via Vulnerability-Preserving Data Augmentation." *Proceedings of the ACM SIGPLAN/SIGBED*

International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES) , 2024, pp. 166–177. [CCF-B / CORE A]

[6] Xiaofei Xie, Wenbo Guo, Lei Ma, Wei Le**, Jian Wang**, Linjun Zhou, Yang Liu, and Xinyu Xing. "Automatic RNN Repair via Model-based Analysis." *Proceedings of the 38th International Conference on Machine Learning (ICML)* , 2021. [CCF-A / CORE A*]

[7] Qing Guo, Felix Juefei-Xu, Xiaofei Xie, Lei Ma**, Jian Wang**, Bing Yu, Wei Feng, and Yang Liu. "Watch Out! Motion is Blurring the Vision of Your Deep Neural Networks." *Advances in Neural Information Processing Systems (NeurIPS)* , 2020. [CCF-A / CORE A*]

[8] Run Wang, Felix Juefei-Xu, Lei Ma, Xiaofei Xie, Yihao Huang**, Jian Wang**, and Yang Liu. "FakeSpotter: A Simple yet Robust Baseline for Spotting AI-Synthesized Fake Faces." *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)* , 2020. [CCF-A / CORE A*]

Peer-Reviewed Journal Articles

[9] Tianlin Li, Xiaofei Xie**, Jian Wang**, Qing Guo, Aishan Liu, Lei Ma, and Yang Liu. "Faire: Repairing Fairness of Neural Networks via Neuron Condition Synthesis." *ACM Transactions on Software Engineering and Methodology (TOSEM)* , vol. 33, no. 1, pp. 1–24, 2023. [CCF-A / CORE A*]

[10] Xiaofei Xie, Tianlin Li**, Jian Wang**, Lei Ma, Qing Guo, Felix Juefei-Xu, and Yang Liu. "NPC: Neuron Path Coverage via Characterizing Decision Logic of Deep Neural Networks." *ACM Transactions on Software Engineering and Methodology (TOSEM)* , vol. 31, no. 3, art. 47, 2022. [CCF-A / CORE A*]

RESEARCH EXPERIENCE

Research Assistant — Singapore Management University (SMU), Singapore *Aug. 2023 – Present*

- Led three first-author projects: Defects4C (LLM repair benchmark for C/C++, ASE'25), execution-trace-augmented code-LLM reasoning (EMNLP-Findings'25), systematic evaluation of AIGC detectors on source code (ASE'24).
- Proposed RATCHET (ISSRE'24), a retrieval-augmented Transformer for program repair, integrating neural and symbolic methods for real-world bug fixing.
- Collaborated with industry partners on reliability of safety-critical software; mentored visiting students and junior PhDs on benchmark design and evaluation methodology.

Research Assistant — Nanyang Technological University (NTU), Singapore *Dec 2019 - Aug 2023*

- Research on testing and security of deep-learning systems: neural-network fairness repair (TOSEM'23), neuron path coverage (TOSEM'22), RNN repair (ICML'21), adversarial robustness (NeurIPS'20), synthetic-media detection (IJCAI'20).
- Core member, AI Singapore Trusted Media Challenge team; designed a hybrid detector for deepfake faces, synthetic audio, and lip-sync inconsistencies; 3rd place among 470 international teams.

INDUSTRY EXPERIENCE

Researcher — Xiaomi AI Lab, Beijing, China *2017 – Dec. 2019*

- Developed a proprietary multi-layer background segmentation and bokeh algorithm (GCN/GAN-based real-time portrait segmentation with multi-depth composition), shipped on Redmi series and select Xiaomi 9 models; contributed to product optimization for the Indian market.
- The algorithm replaced the third-party Meitu SDK, saving ~USD 400K/year in licensing fees and strengthening the portrait-mode product competitiveness.

Backend Engineer / Tech Lead — 58.com Inc., Beijing, China *2011 – 2017*

- Led a 7-person engineering team for the MobileWeb business line serving 15M+ MAU—a major revenue stream; led mobile-app middleware development supporting the company's mobile business across Android, iOS, and MobileWeb.

HONORS & AWARDS

- **AI Singapore Trusted Media Challenge — 3rd Prize Worldwide (3 / 470 teams), 2022**

  Global competition with 470 teams; team designed a hybrid AI model detecting deepfake faces, synthetic audio, and lip-sync inconsistencies—state-of-the-art on Asia-Pacific demographic datasets. Received in-person recognition by Mr Tan Kiat How, then Minister of State for Communications and Information, Singapore. Cumulative prize award: **~USD 82K** (personal: ~USD 14K; seed funding into co-founded startup VAISION: ~USD 68K).

TRANSLATIONAL IMPACT & INDUSTRY PARTNERSHIPS

As technical lead, deployed research outputs across multiple Singapore government agencies, national programs, and industry partners:

- **Government partners:** Research outputs adopted by **DSO National Laboratories** and the **Cyber Security Agency (CSA) of Singapore**; delivered invited technical briefings to CSA and the Ministry of Communications and Information (MCI).
- **National program — Trustworthy AI Centre (TAICeN)**: Led the **AI-Generated Code Detection** thrust; produced the first systematic evaluation framework for AIGC code detectors (ASE 2024, first-author), establishing a detection baseline for Singapore's trustworthy AI-code ecosystem.
- **National program — CREATE (Healthcare Software Security, 100M+ tier)**: Led the **Automated Program Repair** thrust for medical-system code, corresponding to RATCHET (ISSRE 2024, first-author) and Defects4C (ASE 2025, first-author); delivers a repair baseline for safety-critical medical software.
- **Industry partner — MetaTrust**: Deployed the AI-generated code detection technology (ASE 2024, first-author) for **code provenance verification and training-data quality control**, identifying and filtering AI-generated code contamination in training corpora and improving enterprise code-asset trustworthiness.
- **Startup incubation:** Co-founded **VAISION** (trustworthy AI & code-security venture); leveraging the AISG 3rd-place result, **~USD 68K seed capital** injected directly by the challenge organizer.

FUNDED PROJECTS (AWARDS SUPPORTING PUBLICATIONS)

Contributed as student researcher to the following competitive grants:

- **CRPO-GC1-NUS-001** — NRF & CSA, Singapore (CyberSG R&D Programme); *2024–2027*
- **NCRP25-P04-TAICeN** — NRF & CSA, Singapore (National Cybersecurity R&D Programme); *2023–2025*
- **I2301E0026** — RIE2025 IAF-ICP, A*STAR + Alibaba Group, Singapore; *2021–2023*
- **RG12/23** — MOE Academic Research Fund Tier 1, Singapore; *2019–2022*

TALKS & PROFESSIONAL SERVICE

Invited Talks

- **ASE 2025** — Seoul, South Korea.
- **EMNLP 2025 Findings** — Suzhou, China.
- **ISSRE 2024** — Tsukuba, Japan.

Professional Service

- **Registration Chair:** ICECCS 2025, ATVA 2023, APSEC 2020, Internetware 2020 (all CCF-C).
- **External Reviewer:** NeurIPS.

TECHNICAL SKILLS

Programming & ML Frameworks: Python, C/C++, JavaScript, Shell; PyTorch, vLLM.

Program Analysis & Systems: LLVM/Clang, Tree-sitter, GDB-based dynamic tracing; Docker, Linux, Git, distributed training (SLURM + 8×H200), RESTful backend services.

Languages: Mandarin (native); English (fluent research / teaching).

REFERENCES

Available upon request. Suggested referees: Ph.D. advisor Prof. Yi Li (NTU), primary collaborator Prof. Xiaofei Xie (SMU), Prof. Yang Liu (NTU), Prof. Tianwei Zhang (NTU), and Prof. Lei Ma (University of Tokyo).